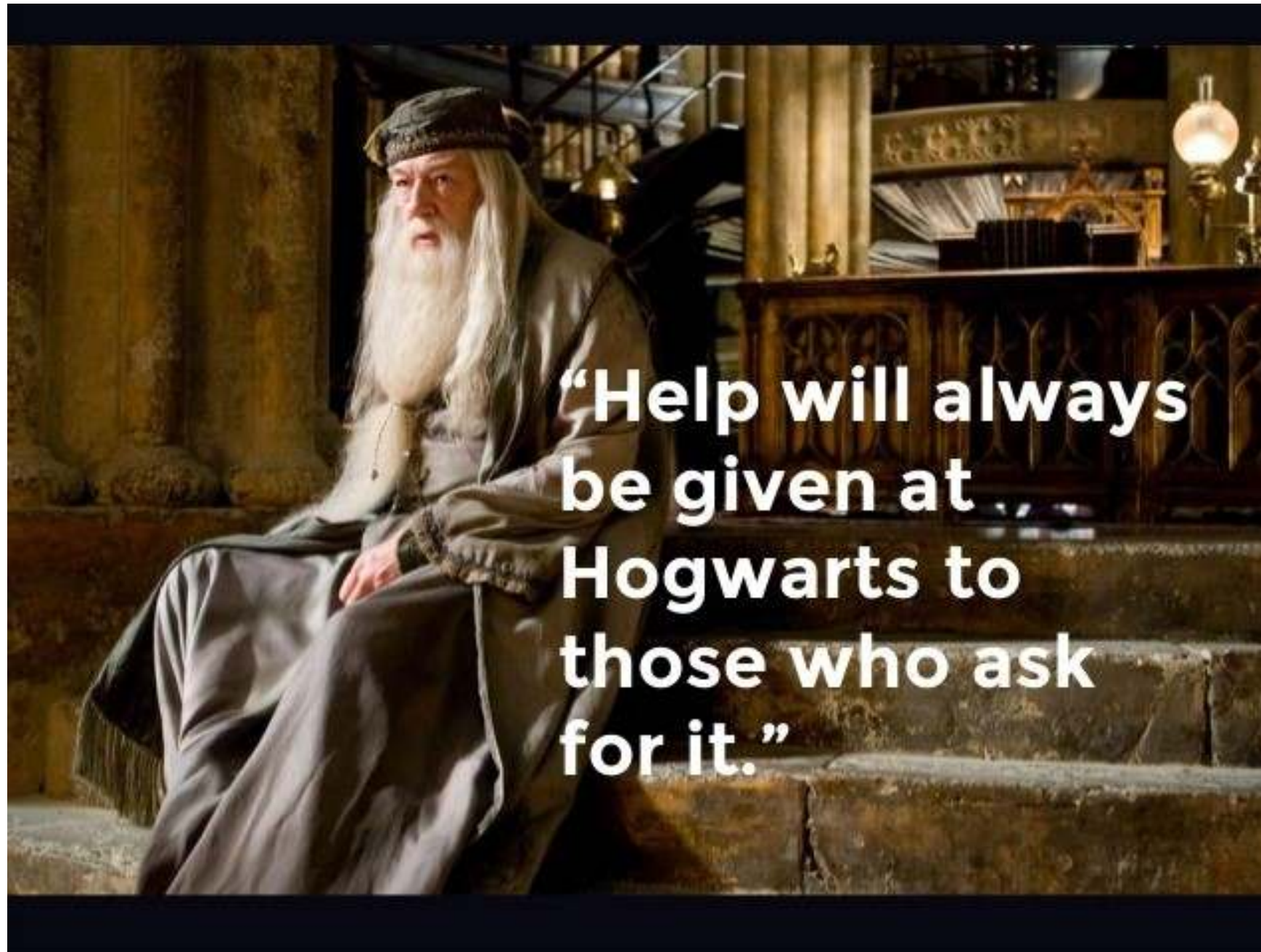


chunkIE: a MATLAB integral equation toolbox

MATLAB: tips and tricks



'help <function name>', 'help <classname>', 'help <classname>/<function name>' provide useful information about how to use well-documented codes in MATLAB

MATLAB: tips and tricks

Don't write buggy codes!



But if you do, and when in doubt, just plot: Printing and plotting are the most efficient ways of debugging your code

MATLAB: tips and tricks

Dot doesn't dot, but not dot dots

Be mucho careful with MATLAB's pointwise multiplication (.*) vs dot product (*)

Speaker Design

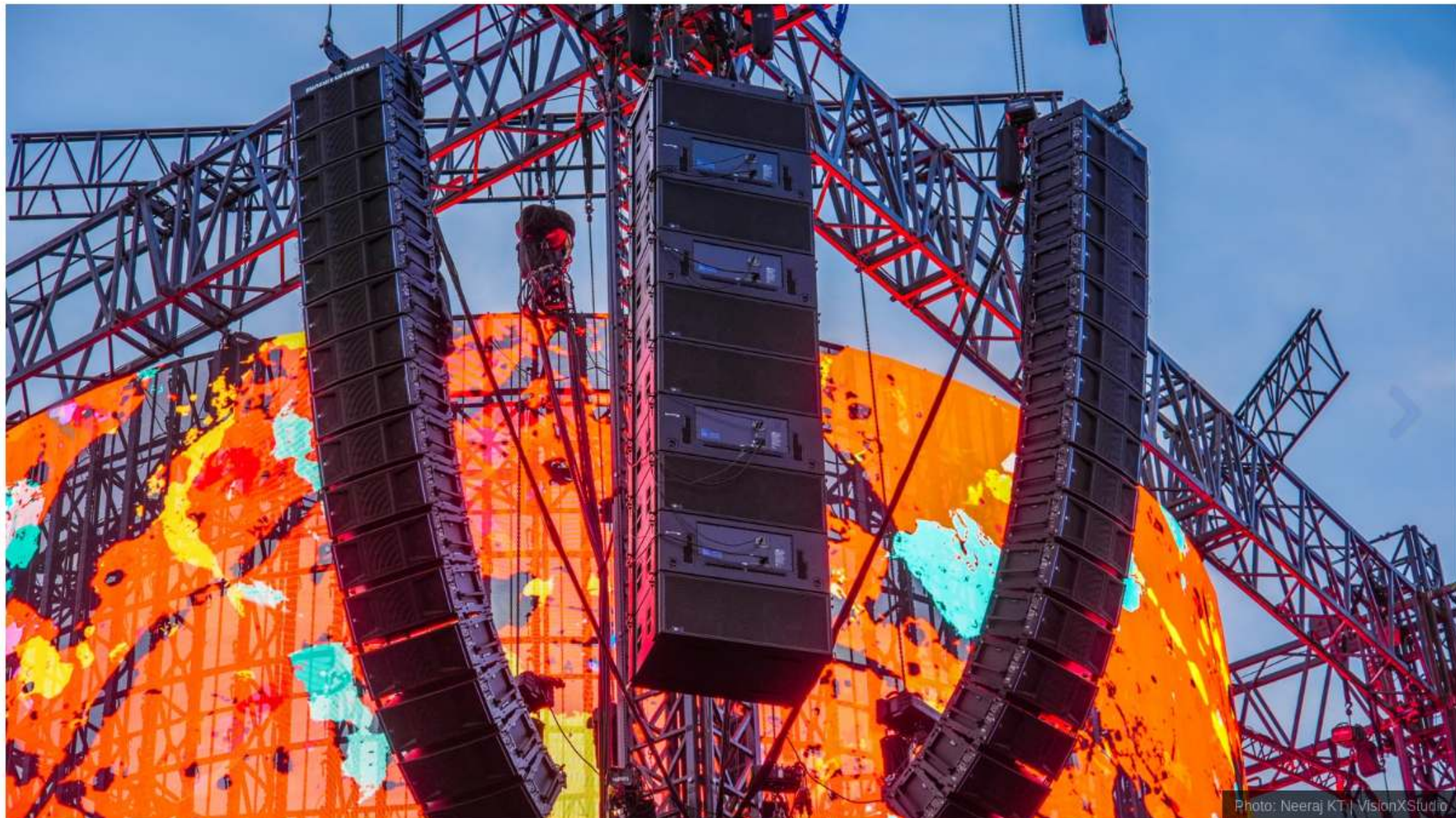


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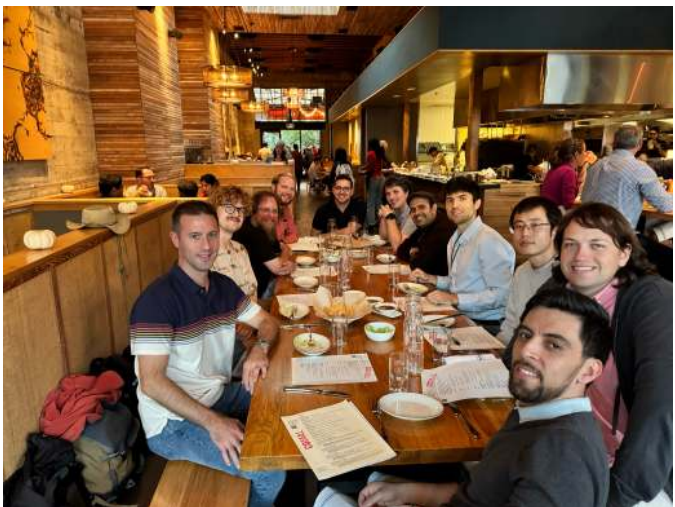
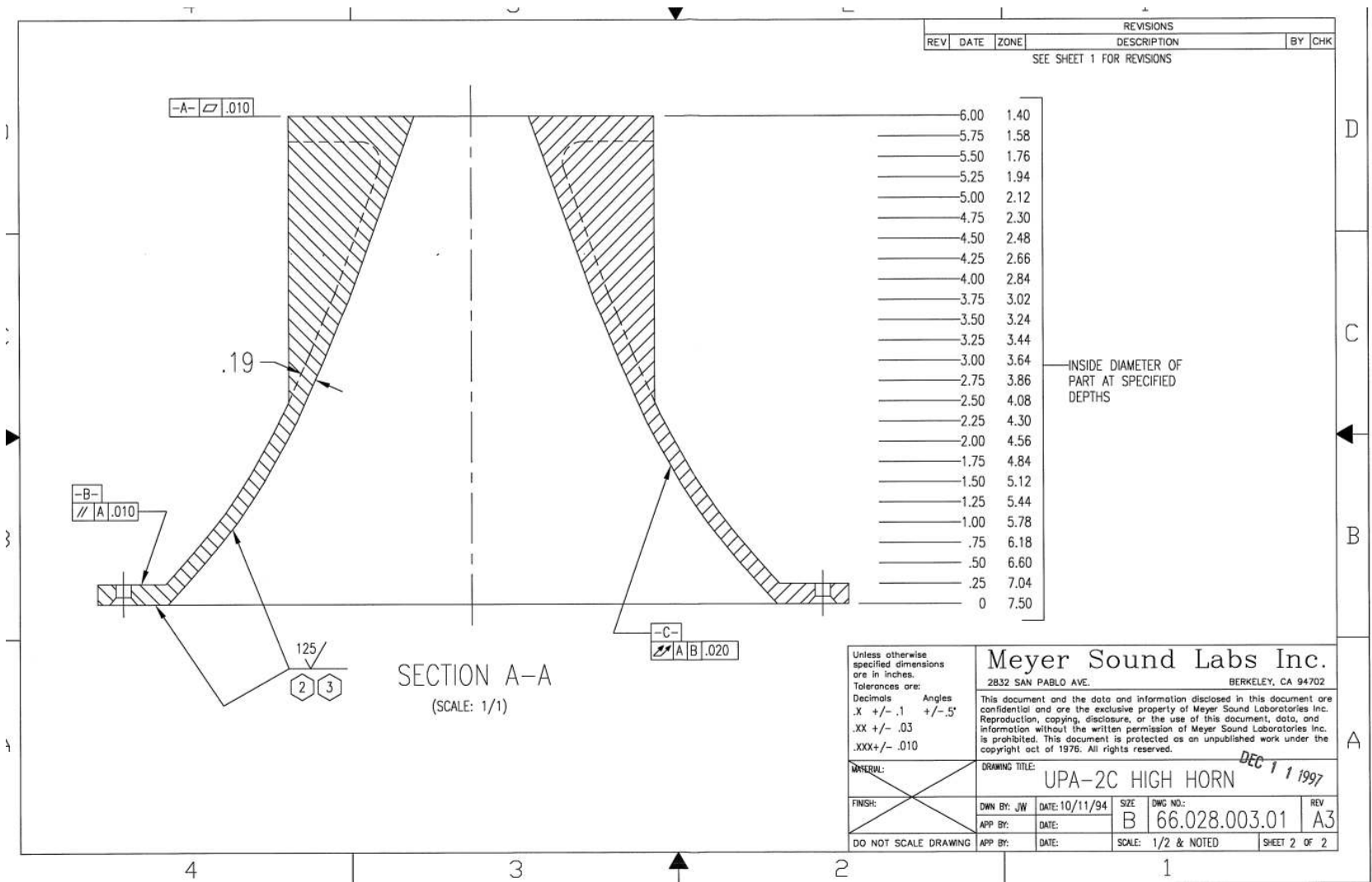
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Ed Sheeran Makes Magic in Mumbai with Meyer Sound

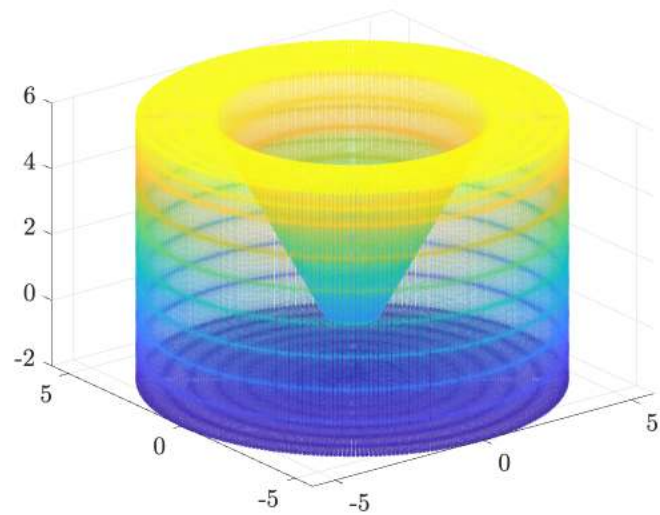
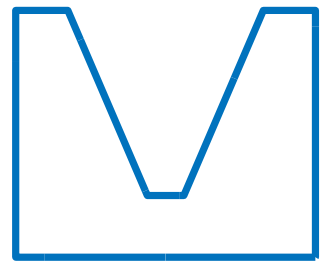
Asia Tour Finale Showcases Seamless Stadium Sound in the Round



Horn loaded speakers



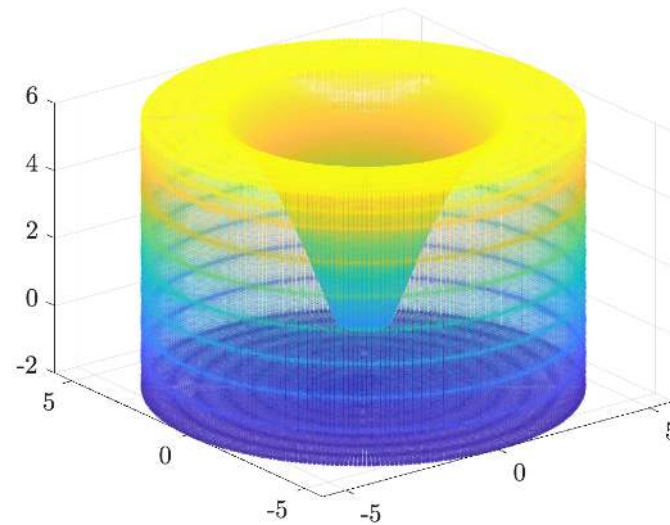
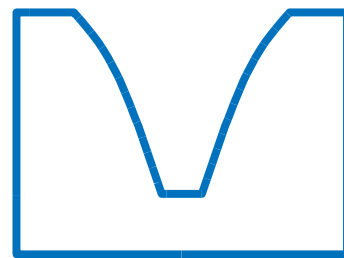
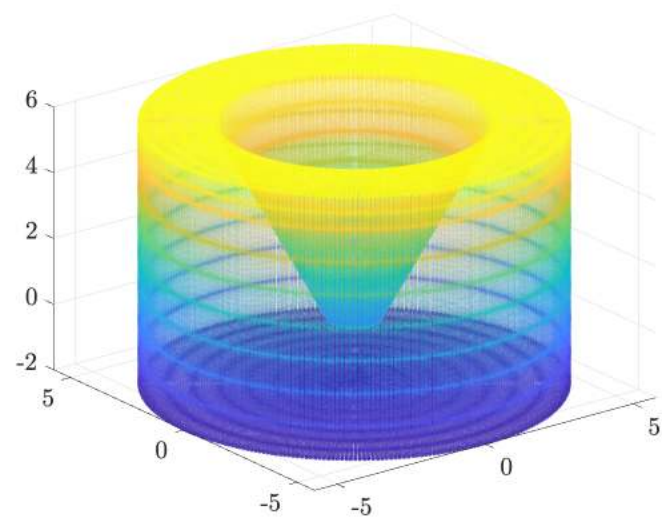
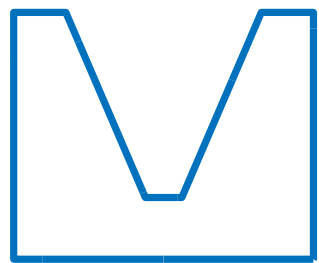
Horn loaded speakers



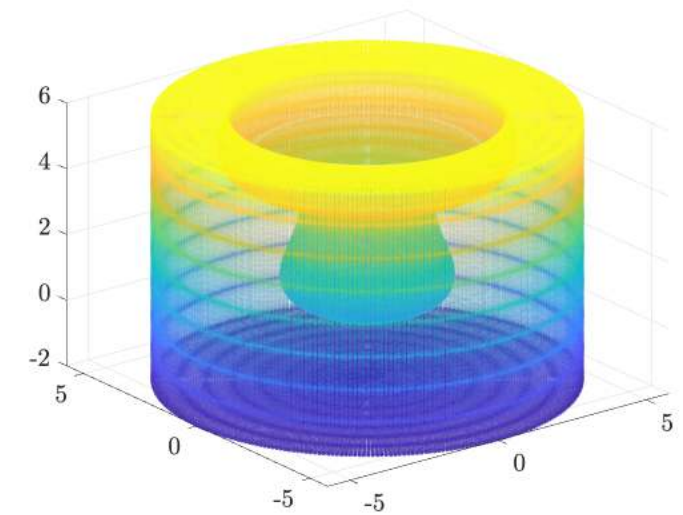
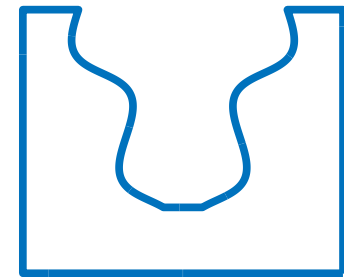
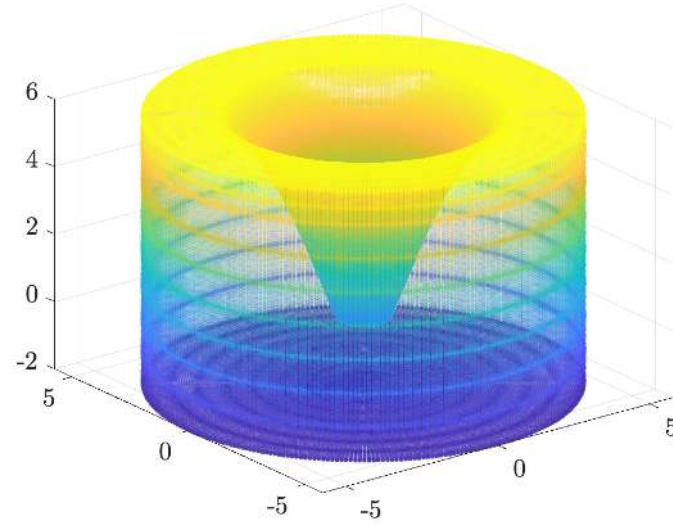
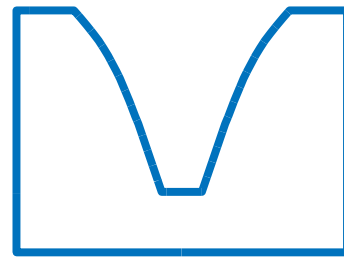
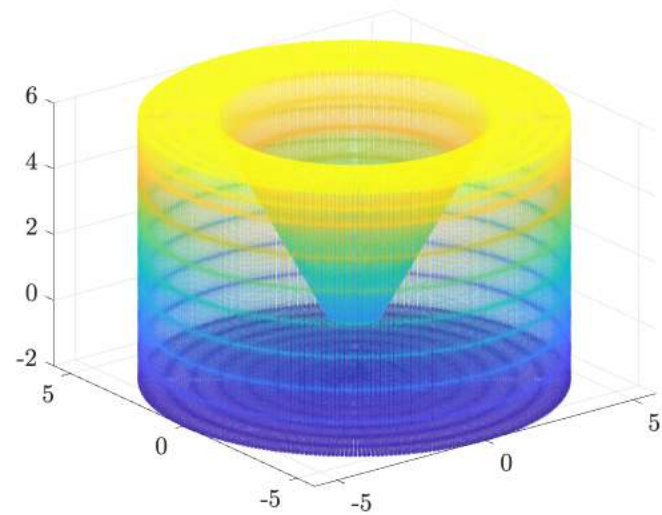
$$(\Delta + k^2)u = 0$$
$$n \cdot \nabla u = f$$

Horn loaded speakers

$$(\Delta + k^2)u = 0$$
$$n \cdot \nabla u = f$$

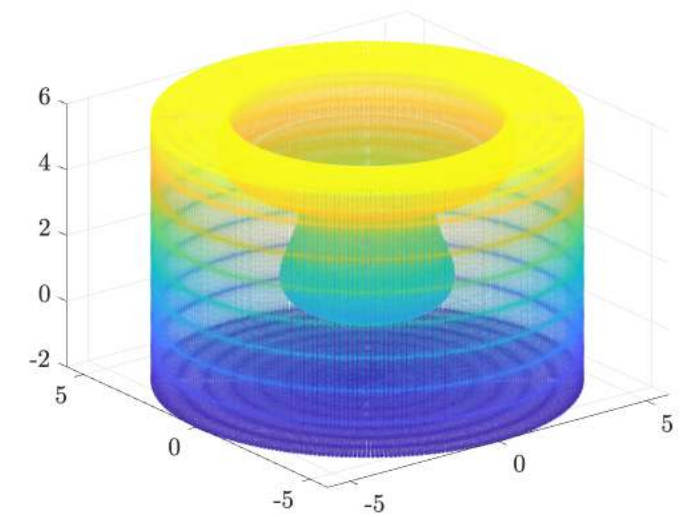
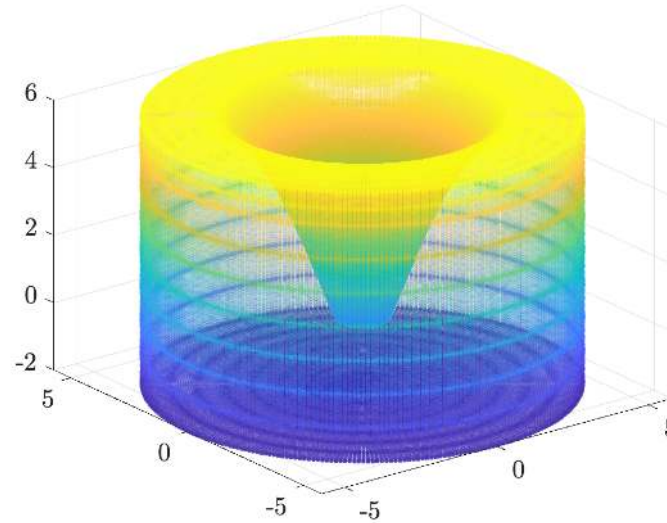
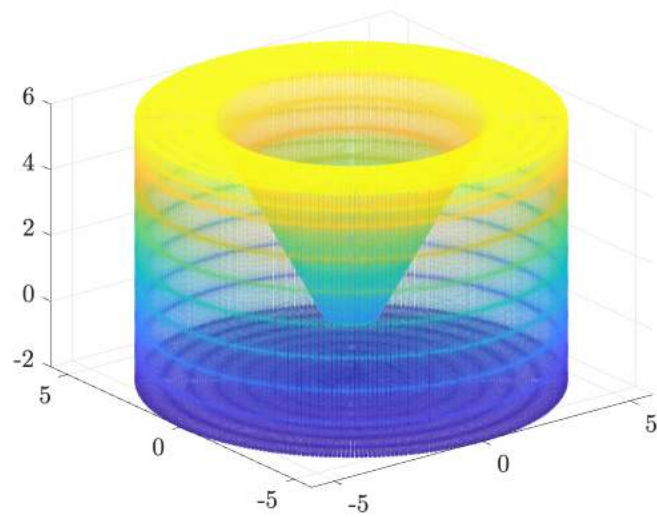
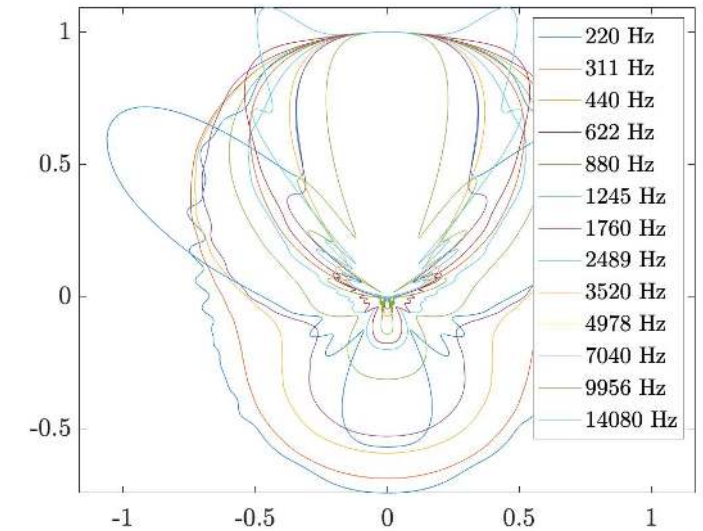
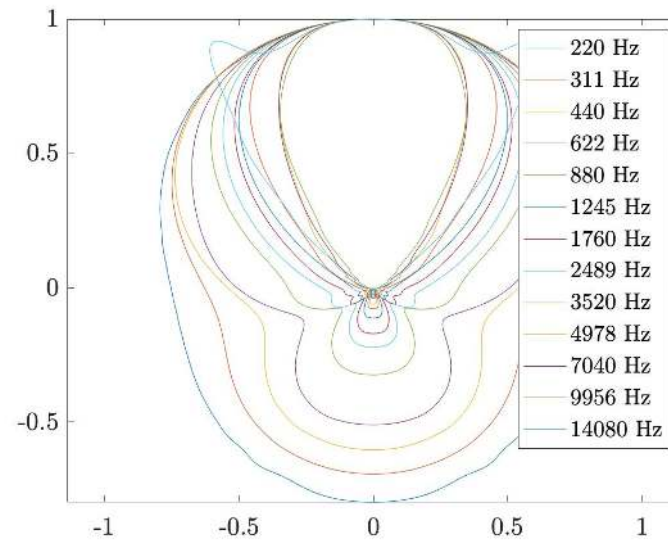
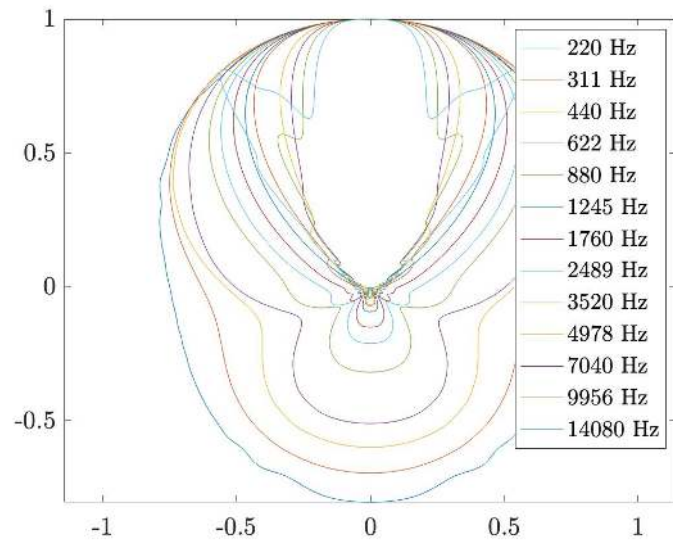


Horn loaded speakers

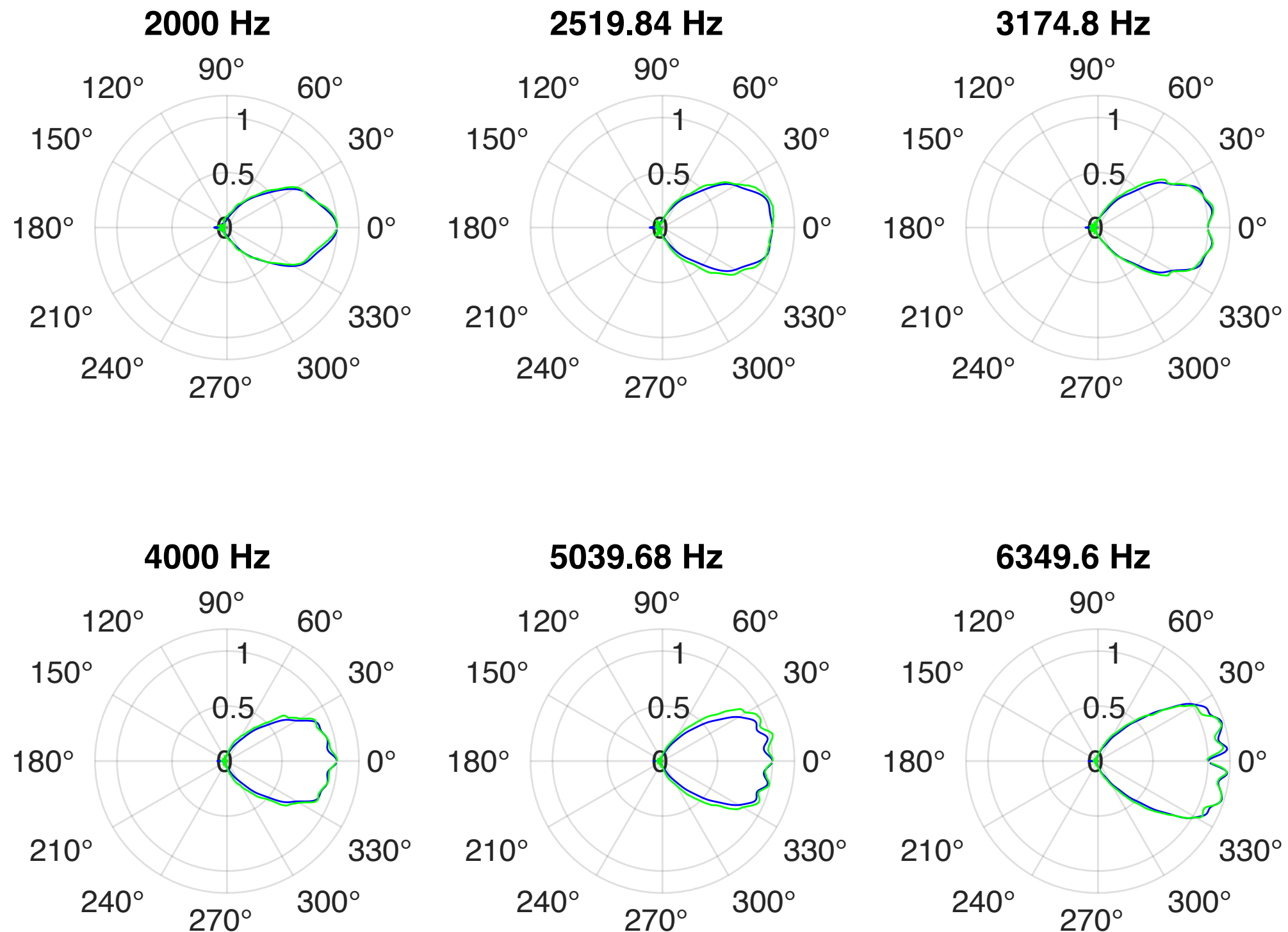
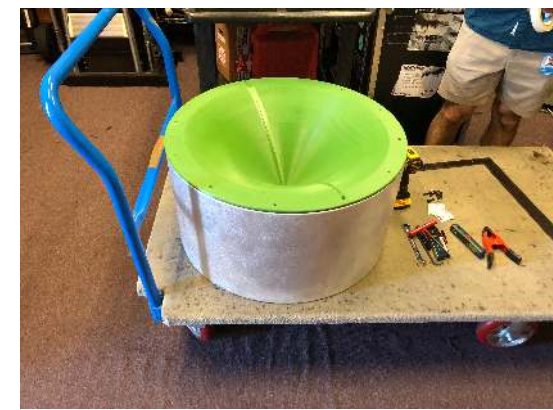


Horn loaded speakers

$$(\Delta + k^2)u = 0$$
$$n \cdot \nabla u = f$$

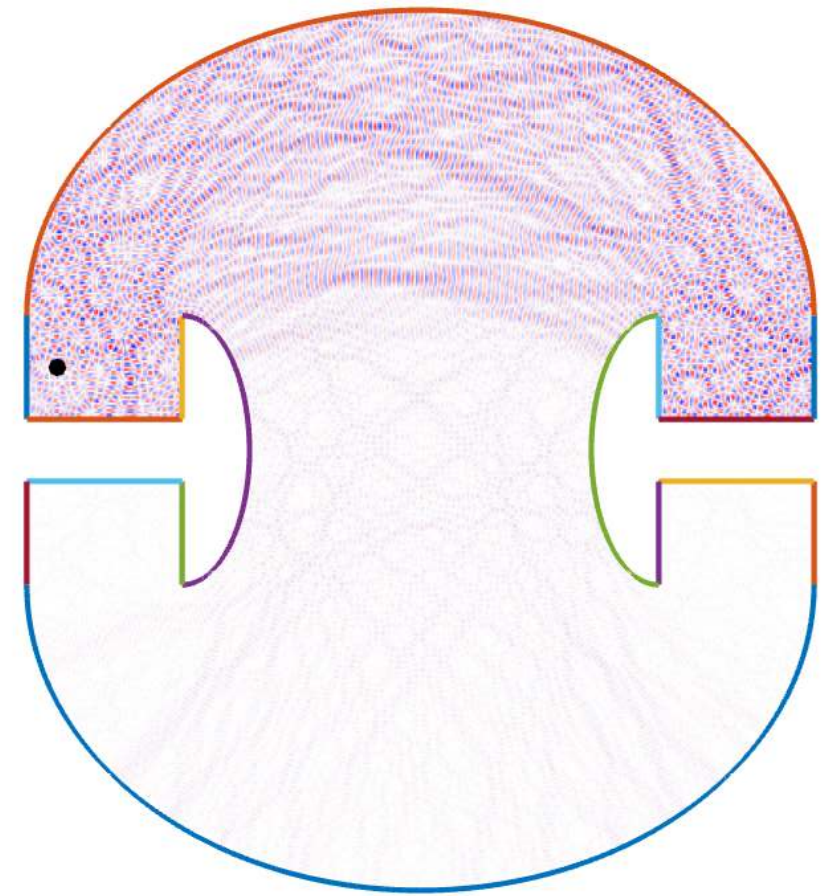


Speaker design and modeling



6 step to live the good life through integral equations

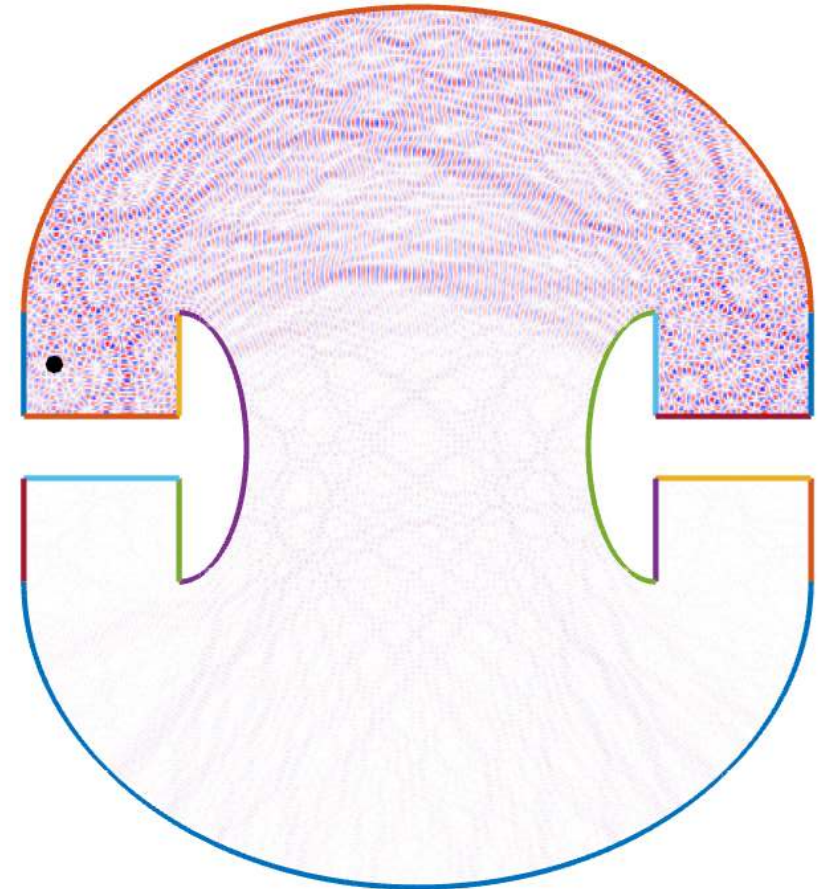
1. Identify a PDE and boundary conditions
2. Convert it into an integral equation
3. Discretize geometry
4. Discretize integral operators
5. Solve the integral equation
6. Post-process/recover PDE solution



6 step to live the good life through integral equations

1. Identify a PDE and boundary conditions

$$\begin{aligned}(\Delta + k^2)u &= 0 && \text{in } \Omega \\ u &= f && \text{on } \partial\Omega\end{aligned}$$



$$\begin{aligned}(\Delta + k^2)u &= 0 && \text{in } \Omega \\ u &= f && \text{on } \partial\Omega\end{aligned}$$

6 step to live the good life through integral equations

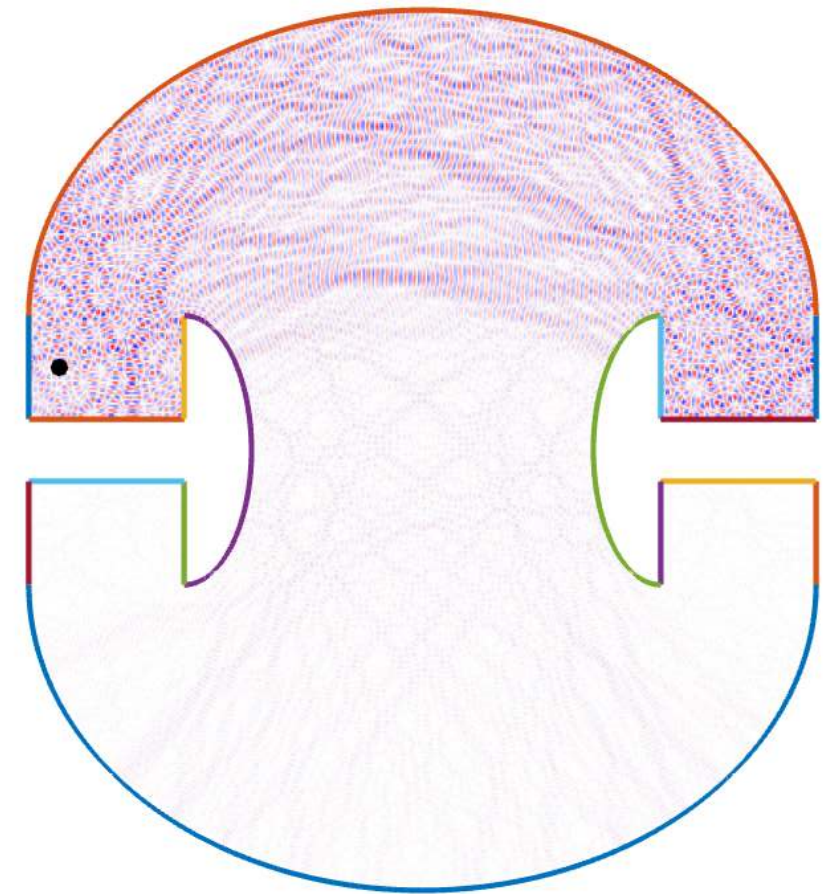
1. Identify a PDE and boundary conditions

2. Convert it into an integral equation

- Integral representations use Green's functions/ fundamental solutions to represent the solution

$$(\Delta_x + k^2)G(x, y) = -\delta_{x=y}$$

$$G(x, y) = \frac{i}{4}H_0^{(1)}(k|x-y|)$$

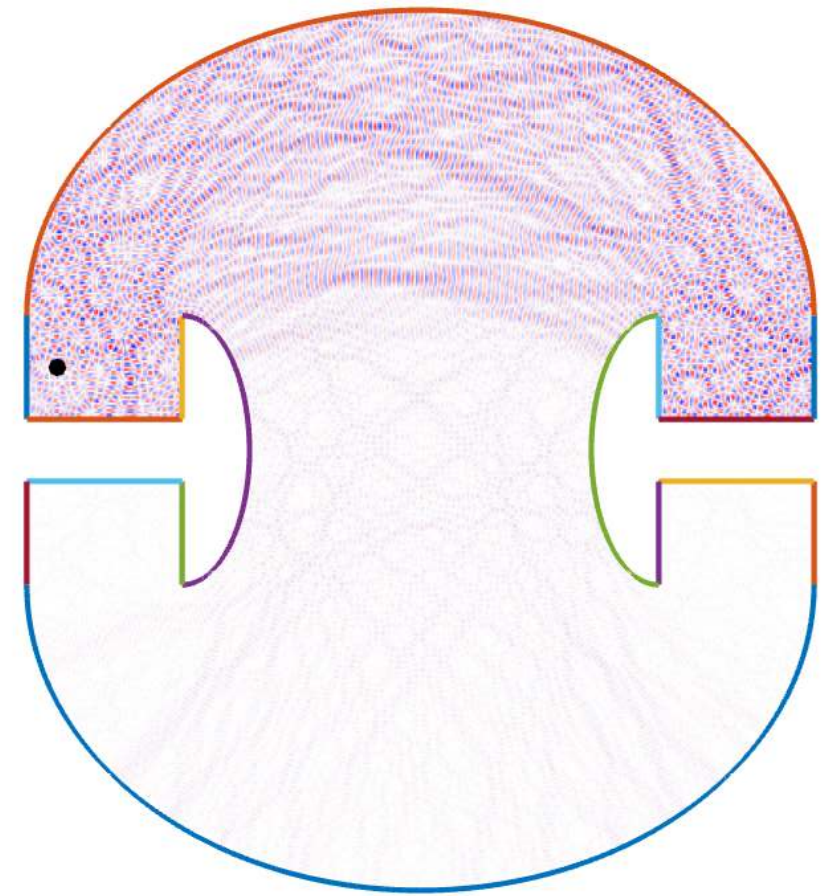


- Information lives on the boundary alone, construct solution from bunch of Green's functions located on the boundary, i.e. set $u(x) = \int_{\partial\Omega} n(y) \cdot \nabla_y G(x, y) \sigma(y) ds_y$ for $x \in \Omega$

- Need to solve for σ such that $-\sigma(x)/2 + \int_{\partial\Omega} n(y) \cdot \nabla_y G(x, y) \sigma(y) = f(x)$ for $x \in \partial\Omega$

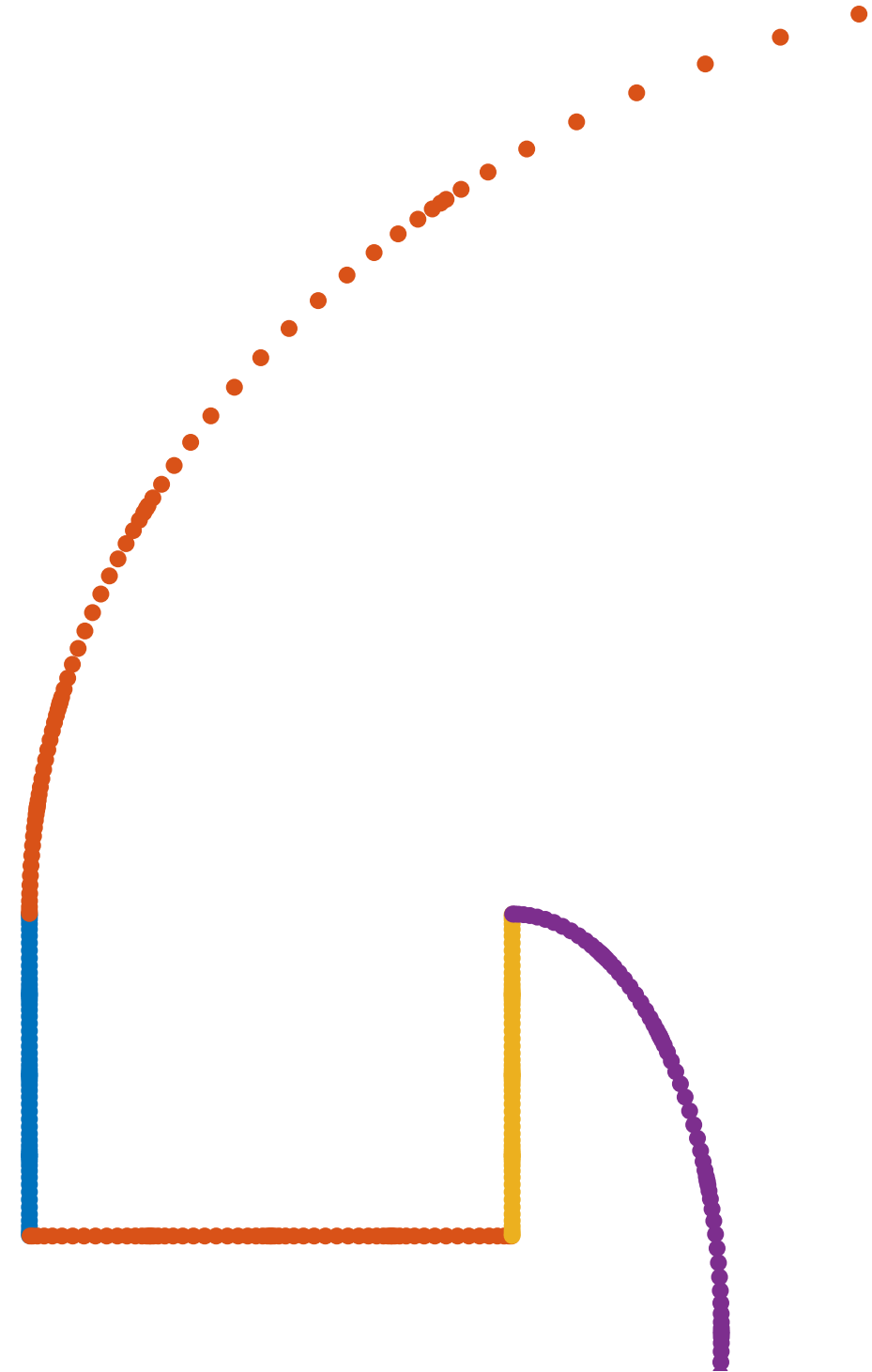
6 step to live the good life through integral equations

1. Identify a PDE and boundary conditions
2. Convert it into an integral equation
- 3. Discretize geometry**
 - Need a way to represent functions and integral equations on the computer, thus need to be able to interpolate functions from finite number of samples, compute their integrals...



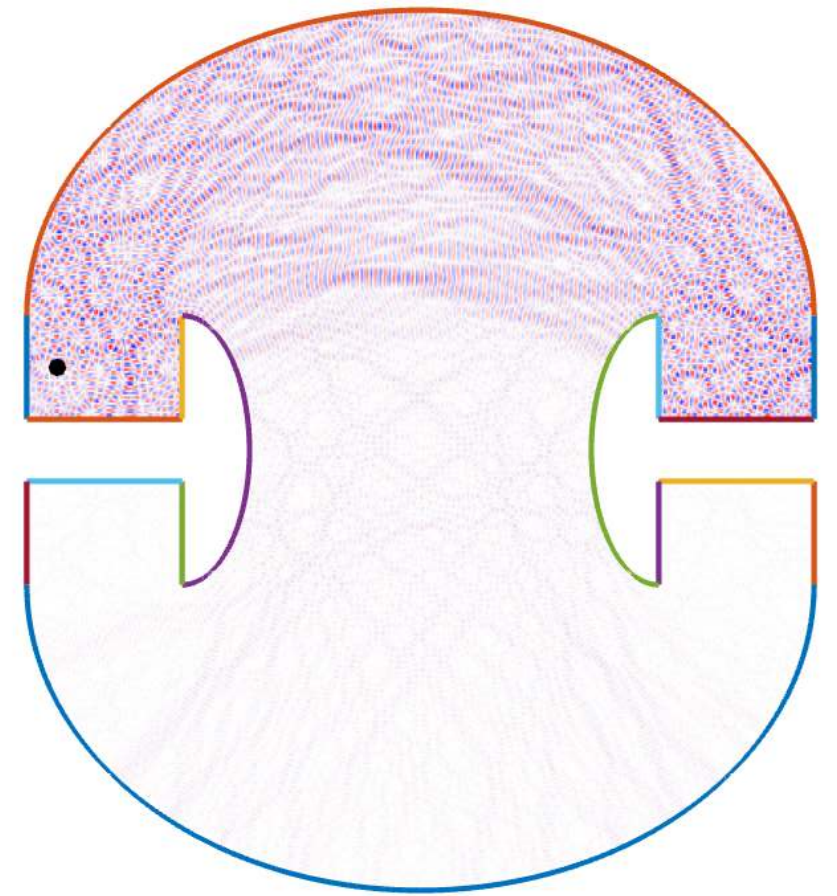
6 step to live the good life through integral equations

1. Identify a PDE and boundary conditions
2. Convert it into an integral equation
3. Discretize geometry
- 4. Discretize integral operators**
 - Draw a ball around each panel and
 - Standard quadrature when x is outside
 - Special quadrature when x is inside



6 step to live the good life through integral equations

1. Identify a PDE and boundary conditions
2. Convert it into an integral equation
3. Discretize geometry
4. Discretize integral operators
- 5. Solve the integral equation**
 - Small systems: backslash/ direct inversion
 - Large systems: GMRES accelerated by FMMs/DMKs (under construction)
 - A fast direct solver



6 step to live the good life through integral equations

1. Identify a PDE and boundary conditions
2. Convert it into an integral equation
3. Discretize geometry
4. Discretize integral operators
5. Solve the integral equation
- 6. Post-process/recover PDE solution**
 - Evaluate $u(x) = D_k[\sigma](x)$

